

1. Administrative & Study Identification

Full Study Title: Lenvatinib (E7080/MK-7902) in Combination With Pembrolizumab (MK-3475) vs. Standard Chemotherapy and Lenvatinib Monotherapy in Participants With Recurrent/Metastatic Head and Neck Squamous Cell Carcinoma That Progressed After Platinum Therapy and Immunotherapy (MK-7902-009/E7080-G000-228/LEAP-009)

Official Title: A Phase 2, randomized, open-label three-arm clinical study to evaluate the safety and efficacy of lenvatinib (E7080/MK-7902) in combination with pembrolizumab (MK-3475) versus standard of care chemotherapy and lenvatinib monotherapy in participants with recurrent/metastatic head and neck squamous cell carcinoma (R/M HNSCC) that have progressed after platinum therapy and immunotherapy (PD-1/PD-L1 inhibitors) **Short Title/Acronym:** LEAP-009 **Posting ID:** 503779518131649734 **Protocol Number:** MK-7902-009 / E7080-G000-228 **NCT Number:** NCT04428151 **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Sponsor / Company:** Merck Sharp & Dohme LLC (MSD) / Merck **Investigational Product:** Lenvatinib (LENVIMA®) and Pembrolizumab (KEYTRUDA®) **Phase of Development:** Phase II **Medical Monitor:** MSD Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the Overall Survival (OS) of lenvatinib in combination with pembrolizumab versus standard-of-care chemotherapy in patients with R/M HNSCC. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Objective Response Rate (ORR), Duration of Response (DOR), and to assess the safety and tolerability of the treatment regimens.

2.2 Background & Rationale Researchers are seeking new ways to treat patients with recurrent or metastatic head and neck cancer whose disease has progressed after treatment with platinum-based therapy and immunotherapy (PD-1/L1 inhibitors). Standard-of-care options for these patients often have limited efficacy. The goal of this study is to learn if more people who receive lenvatinib and pembrolizumab have a better overall survival rate than people who receive standard chemotherapy treatment.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Standard-of-care chemotherapy) **Blinding:** Open-label **Randomization:** Randomized (3:3:2 ratio), stratified by PD-

L1 tumor proportion score (<50% vs ≥50%) and ECOG PS (0 vs 1).

Protocol Amendment: With Amendment 7, participants will discontinue lenvatinib and pembrolizumab and lenvatinib monotherapy, unless discussed with the Sponsor.

3.2 Planned Enrollment & Duration Target **\$N\$:** 408 participants (Targeted Enrollment) **Number of Sites:** Multicenter (North America, Europe, Asia, and Australia) **Estimated Study Start:** 06-Aug-2020 **Primary Completion:** 31-May-2024 (A protocol-specified periodic safety review was completed with a data cut-off of 31-May-2024. Per protocol, 34 participants enrolled after the primary completion date and will be analyzed in the End of Trial analysis.)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years (Male and Female). 2. Pathologically confirmed recurrent (incurable) or metastatic head and neck squamous cell carcinoma (HNSCC) of the oral cavity, oropharynx, hypopharynx, and/or larynx. 3. Documented disease progression on or after treatment with a platinum-containing (e.g., carboplatin or cisplatin) regimen and a programmed cell death protein 1/programmed death-ligand 1 monoclonal antibody (anti-PD-1/PD-L1 mAb). 4. Hepatitis B (HBsAg positive) participants receiving antiviral therapy > 4 weeks with undetectable viral load, and Hepatitis C (HCV) participants with undetectable viral load are eligible. 5. Adequately controlled blood pressure (BP). 6. Compliance with specific male and female contraceptive guidelines across treatment arms.

4.2 Exclusion Criteria 1. Disease that is suitable for local therapy administered with curative intent. 2. History of (noninfectious) pneumonitis/interstitial lung disease that required steroids, active tuberculosis, known HIV infection, or allogeneic tissue/solid organ transplant. 3. Life expectancy of < 3 months, rapidly progressing disease, or clinically significant cardiovascular impairment within 12 months prior to the first dose. 4. Previously treated with 4 or more systemic regimens given for recurrent/metastatic disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm 1:** Oral lenvatinib (20 mg daily) + Intravenous pembrolizumab (200 mg every 3 weeks for up to 35 cycles). * **Arm 2:** Standard-of-care (SOC) chemotherapy (Investigator's choice of docetaxel, cetuximab, capecitabine, or **Paclitaxel** [ATC Code: L01CD01; Trade Names: Abraxane, Onxol, Taxol, Paccal; EMA Info: https://www.ema.europa.eu/en/documents/product-information/abraxane-epar-product-information_en.pdf]). * **Arm 3:** Oral lenvatinib monotherapy (24 mg daily). **Route of Administration:** Oral

(lenvatinib, capecitabine) and Intravenous (pembrolizumab, docetaxel, paclitaxel, cetuximab). **Duration of Treatment:** Until centrally verified disease progression, unacceptable toxicity, or withdrawal (up to a maximum of 35 cycles for pembrolizumab).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing treatment-related toxicities. **Prohibited:** Prior treatment with lenvatinib; live or live-attenuated vaccines within 30 days prior to the first dose; systemic steroid therapy (>10 mg daily prednisone equivalent) within 7 days prior to the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Imaging, ECOG PS, PD-L1 Testing
Baseline	Day 0	Randomization, First Dose
Interim	Every 3 Weeks	Treatment administration, Safety Labs, AE tracking. Tumor imaging every 6 weeks through Year 1, then every 9 weeks
End of Study	Up to 4 Years	Safety follow-up 30 days post-treatment (90 days for SAEs), Long-term OS monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Survival (OS). OS is defined as the time from randomization to death due to any cause.

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) by BICR. * Objective Response Rate (ORR). * Duration of Response (DOR). * Safety and tolerability (Number of participants experiencing AEs/SAEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: AEs are monitored continuously throughout the study and for 30 days following the end of treatment. Graded according to NCI CTCAE v5.0. **Serious Adverse Events (SAEs):** Reported promptly and monitored for up to 90 days following treatment end. **Data Monitoring Committee (DMC):** An external Data Monitoring Committee (eDMC) conducts protocol-specified periodic safety reviews to assess the ongoing safety and risk/benefit ratio of the investigational regimens.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary and secondary efficacy outcome measures; Safety Set for all AE/SAE

evaluations. **Sample Size Justification:** A targeted enrollment of 408 participants, randomized 3:3:2 to adequately power the primary endpoint (OS) and secondary endpoints (PFS, ORR). **Handling of Missing Data:** Participants without documented progression or death are censored at the date of their last evaluable imaging assessment for time-to-event analyses (PFS/OS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. Phase requirement matched for Phase 2 standards. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence. **Audit Trail:** Secure Electronic Data Capture (EDC) systems featuring lock mechanisms, electronic data clarification forms, and comprehensive audit trails.

11. Study Identifiers & Governance

Primary ID: MK-7902-009 **Registry Number:** NCT04428151 **Posting ID:** 503779518131649734 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC / Merck **Study Title:** LEAP-009 **Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting)

12. Clinical Framework (Oncology / HNSCC Specific)

Disease Area: Squamous Cell Carcinoma of Head and Neck **Preferred UMLS Name:** Squamous cell carcinoma of the head and neck **MeSH Code:** D000077195 | **SNOMED CT ID:** 716659002 **Semantic Type:** Neoplastic Process **Disease Definition:** The most common type of head and neck carcinoma that originates from cells on the surface of the NASAL CAVITY; MOUTH; PARANASAL SINUSES, SALIVARY GLANDS, and LARYNX. Mutations in TNFRSF10B, PTEN, and ING1 genes are associated with this cancer. **Diagnosis Threshold:** Tumor progression post-platinum therapy and PD-1/L1 inhibitor **Medical Methodology:** Tyrosine Kinase Inhibitor (TKI) + Immune Checkpoint Inhibitor Combination Therapy vs. Standard Cytotoxic Chemotherapy **Optimization Target:** Improvement in Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for R/M HNSCC **Education Protocol (HCP):** Investigator meetings and site initiation visits detailing modified RECIST 1.1 criteria **Education Protocol (Patient):** Informed consent, oral medication dosing diaries, and AE self-reporting guidance **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for impartial and accurate radiologic assessments

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 4 years **Onsite Visit Cadence:** Every 3 weeks aligned with treatment cycles **Remote Monitoring Frequency:** Follow-up contacts for survival status and safety as required **Checklist Review Interval:** Imaging assessments every 6 weeks through Year 1, and every 9 weeks thereafter

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				